

Data Quality Plan

1. Introduction

This Data Quality Plan intends to assess a subset of data collected by the Residential Property Price Register (RPPR). Within the sub-dataset, there are nine features (*DateofSale(dd/mm/yyyy)*, *Address*, *County*, *Eircode*, *Price*, *NotFullMarketPrice*, *VATExclusive*, *DescriptionofProperty*, *PropertySizeDescription*) describing performance of residential properties purchase in Ireland from 2016 to 2024. This Data Quality Plan will provide detailed analysis on each feature, including: list the data quality issues, proposing potential solutions to these issues, and justifying the selected solutions. This plan aims to produce a final cleaned dataset with clearly defined feature types.

2. Findings and Discussion

In this part of the Data Quality Plan, we will, by each feature, make a final decision on feature types, discuss identified issues, compare and choose the most appropriate solutions to those. Table 1 shows an overview of our assessment of each feature, which will be elaborated in this part.

2.1 *DateofSale(dd/mm/yyyy)*

The original *DateofSale(dd/mm/yyyy)* feature shows relatively high cardinality. It records the exact transaction date of each sale. On day level, we cannot aggregate trends. Such precision is not helpful for predictive modelling and may introduce noise to later analysis. Therefore, we propose two solutions to handle this high cardinality. First, we extract information on both month and year from the original feature *DateofSale(dd/mm/yyyy)*, and focus on monthly-level analysis afterwards. Second, we group the original feature to quarterly or yearly time period to reduce cardinality.

After comparing the two solutions, we choose to proceed with the first solution: extract information on month and year from the original feature. Monthly aggregation provides a balance between cardinality of feature and sufficient information to observe trends in the residential property market. Compared with the second solution, monthly aggregation allows more precise analysis on trends and avoids losing too much information.

After applying the first solution, we remove the original *DateofSale(dd/mm/yyyy)* column and add *SaleYear* and *SaleMonth* as the result of our data cleaning. *SaleYear* and *SaleMonth* are both treated as categorical variables.

2.2 *Address*

Address feature also shows high cardinality as each transaction is linked to a specific address. Analyzing properties one by one is not robust enough to support meaningful market analysis. The large number of unique address values makes it difficult to identify broader geographic patterns within the dataset. Thus, we have two possible solutions to further discuss. First, we drop *Address* feature due to high cardinality, and use *County* and *Eircode* features together to provide regional information. Second, we initially group *Address* into Dublin and non-Dublin areas. For non-Dublin area, we use *County* feature as location markers. For Dublin region, we extract postal district information from the original *Address* feature.

After comparing the two solutions, we decide to adopt the second approach for the following reason. While dropping the *Address* feature simplifies the dataset, it may lead to loss of potentially valuable location information, particularly within Dublin where property prices may vary significantly across different postal districts. Also, using *County* feature instead, will lose the chance to compare trends and differences within Dublin. *Eircode* feature is reporting a missing rate of nearly 70%, making it hard to provide valuable analysis. Thus, combining *County* and *Eircode* features is not appropriate enough to provide robust research

on location analysis. However, for the second solution, extracting Dublin postal districts and using County for non-Dublin areas, can preserve meaningful geographic granularity while still reducing the high cardinality of the original *Address* feature. This approach improves interpretability and allows more effective analysis of regional property market patterns.

When applying the second solution, we have similar rows as the result of initial extraction, for example, D7 and DUBLIN07, D1 and DUBLIN1. Under Irish context, we recognize that this represents the same district. So, we merge them together. Figure 1 shows the result of our final cleaning. From Figure 1, we have Non-DUBLIN represents non-Dublin areas where detailed location information can be provided by *County* feature. For Dublin area, we have regional information within each postal district, from DUBLIN1 to DUBLIN24, while DUBLIN_OTHER refers to places we cannot extract district number from the original *Address* feature.

After applying the second solution, we setup a new feature *Region* to record the result of our data cleaning. New feature *Region* is viewed as a categorical variable.

2.3 County

The *County* feature is treated as a categorical variable, representing the administrative counties of Ireland. As pointed out in the Data Quality Report, this feature does not present significant data quality issues. There are no missing values, and the cardinality of the feature is 26, which aligns with the official number of counties in Ireland.

Given that geographic location plays an important role in residential property market analysis, this feature provides meaningful regional information for further investigation, especially for non-Dublin areas for the reason discussed above in the 2.2 Address section. Therefore, we decide to keep *County* feature in its original form with no extra transformation needed. Together with new feature *Region*, *County* feature will also serve as an important location indicator.

2.4 Eircode

For feature *Eircode*, we have identified several data quality issues in the Data Quality Report. First, the missing rate of this feature is 68.86%. Such a high proportion of missing values reduces the reliability of analyses based solely on this variable and limits its usefulness in providing location information. Second, the feature has a very high cardinality of 16,761, as each Eircode refers to a specific property location. This level of precision makes it difficult to identify broader regional patterns. Therefore, we consider several potential approaches to address these issues. First, we drop this feature due to high missing rate and high cardinality. Second, we try to fill in the missing part with information from *Address* feature. This may include call external APIs to map address with a certain Eircode.

After further assessing the second solution, we still find certain constraints. Specifically, there is limited open API that can provide the required mapping relationship. Moreover, even if we can fill in nearly 70% of missing data, we still need to aggregate to regional level to support later research, because Eircode refers to one single property. Plus, we have already done the regional aggregation in section 2.2 Address. In this way, we choose the first solution, drop *Eircode* feature, as the most appropriate action to this feature.

2.5 Price

The *Price* feature is finally treated as a continuous variable, representing the transaction value of residential properties. According to the Data Quality Report, the distribution of this feature is highly right-skewed, where most transactions fall within a moderate price range while a small number of observations show extremely high values. These extreme values may be caused by portfolio transactions.

These extreme values can significantly influence statistical analysis and the results of subsequent analysis or modelling. Therefore, we propose two possible solutions targeting at these issues. First, we pick out and drop portfolio transactions according to addresses, and then, we clamp *Price* feature to the upper bound. Second, we apply a log transformation to the *Price* feature, which can help with the skewness and slightly reduce the influence of those extreme values.

After comparing the two approaches, we choose the clamping method. While log transformation can effectively reduce skewness, it also changes the original scale of the *Price* feature. Clamping, on the other hand, directly limits the impact of extreme values while preserving the original monetary scale, making the results easier to interpret and more suitable for further analysis. Also, since *Price* is our target value, keeping its original scale remains top priority. Another reason for choosing clamping is that some extreme values are associated with portfolio transactions, such as the purchase of an entire building or multiple flats together, which is not the focus of our research. Dropping portfolio transactions can help minimize the distorted influence of portfolio transactions.

When applying the clamping method, we follow the IQR-based methodology. To be more specific, from our dataset, we firstly identify 3,009 values higher than the upper bound. Then, we clamp these values to the upper bound. After this, we valid our implementation by checking that the maximum price, now, is equal to the upper bound value. Figure 2 displays price distribution after clamping. From Figure 2, we can see that extreme values above the upper bound have been capped, and the maximum value now equals the predefined upper bound. As a result, the current distribution becomes more stable and suitable for further analysis.

2.6 NotFullMarketPrice

NotFullMarketPrice feature is viewed as a categorical variable with binary values indicating whether a transaction was conducted at full market price. Based on findings from the Data Quality Report, this feature does not indicate significant data quality issues. There are no missing values and the feature contains only two possible categories, which keeps its cardinality low.

Although the majority of transactions occur at full market price, this variable still provides useful information about residential property transactions. Therefore, we decide to keep the *NotFullMarketPrice* feature without extra transformation for further analysis.

2.7 VATExclusive

Feature *VATExclusive* is treated as a categorical variable with binary values indicating whether the recorded transaction price excludes VAT. Based on findings from the Data Quality Report, this feature does not report significant data quality issues. There are no missing values and the feature contains only two possible categories, which keeps its cardinality low.

As VAT may influence the interpretation of property prices, particularly for newly built properties, this feature provides useful information when analysing the *Price* feature. Therefore, we decide to keep the *VATExclusive* feature in its original form for later research.

2.8 DescriptionofProperty

Feature *DescriptionofProperty* is viewed as a categorical variable, as it contains a limited number of predefined property types. According to the Data Quality Report, this feature does not have major data quality issues. There are three types of properties and there are no missing entries, indicating the feature is well-recorded within the dataset.

This feature provides meaningful information about the type of property involved in each transaction, which can be useful when analyzing transaction prices across different property types. Therefore, we decide to keep the *DescriptionofProperty* feature without applying additional transformations.

2.9 PropertySizeDescription

Feature *PropertySizeDescription* is considered as a categorical variable because it contains four size ranges of a property. However, according to the Data Quality Report, this feature reports a significant data quality issue: the missing rate is nearly 95%. Such a large proportion of missing values makes the feature unreliable for meaningful statistical analysis, because conclusions based on the remaining 5% of values cannot represent the overall dataset.

To address this issue, two potential solutions can be considered. First, we drop *PropertySizeDescription* feature from the dataset due to the lack of valid observations. Second, we keep this feature and treat the missing part as a separate category. In this way, the limited available information can be used in later analysis.

After comparing the two solutions, we choose to drop the *PropertySizeDescription* feature. Given the extremely high missing rate and the absence of reliable external data sources to recover the missing values, keeping the feature may introduce more noise than useful information. Dropping it helps maintain the overall data quality and prevents misleading conclusions in later research.

3. Conclusion

This Data Quality Plan evaluates nine features from the Residential Property Price Register (RPPR) dataset and identifies several data quality issues that may affect further analysis. The main issues include high cardinality in *DateofSale(dd/mm/yyyy)* and *Address* features, a high missing rate in *Eircode* and *PropertySizeDescription*, and extreme values in the *Price* feature.

To address these issues, several data cleaning and transformation strategies are applied. The *DateofSale(dd/mm/yyyy)* feature is transformed into *SaleYear* and *SaleMonth* to reduce cardinality and to provide monthly aggregated analysis. Similarly, feature *Address* is extracted into a new *Region* feature to capture geographic patterns, particularly within Dublin. Features with limited analytical value, such as *Eircode* and *PropertySizeDescription*, are dropped due to high missing rates. In addition, extreme values in the *Price* feature are handled via an IQR-based clamping method to reduce the influence of outliers while maintaining the original scale of the variable.

After implementing these data cleaning solutions, the dataset now contains well-defined feature types and improved data consistency. Therefore, the cleaned dataset is more suitable for subsequent statistical analysis and modelling to capture and conclude on residential property market trends in Ireland.

Appendix

A comprehension of data issues

Feature	Type	Data Quality Issue(s)	Impact	Root Cause Analysis	Proposed Solution	Final Action
DateofSale(dd/mm/yyyy)	object	Too much cardinality, not useful for prediction.	High	Natuaral records of dates	Extract year and month	Drop the original column and extract year and month from it.
Address	object	Too much cardinality, not useful for prediction.	High	Natuaral records of different properties	Investigate weather it can be converted to eircode. If not, drop it.	Drop the original column and extract area
County	category	No issue	Low	No issue	Keep	Keep
Eircode	object	Too much cardinality, not useful for prediction.	High	Natuaral records of different properties	Try to fill it by address and extract the district from eircode. If can't, drop it.	Drop
Eircode	object	Too much missing(68.86%), not useful for prediction.	High	Data not collected	Try to fill it by address. If can't, drop it.	Drop
Price	float64	Outliers and extremes	Medium	Most high values are valid outliers but several extremes need more investigation	Clamp the price to upper bound	Pick out and drop the bulk property transactions according to addresses , and clamp the prices of houses exceeding the upper bound to the upper bound.
NotFullMarketPrice	category	No issue	Low	Binary categorical feature	Keep	Keep
VATEExclusive	category	No issue	Low	Binary categorical feature	Keep	Keep
DescriptionofProperty	category	No issue	Low	Records of property types	Keep	Keep
PropertySizeDescription	category	Too much missing(97.73%), not useful for prediction.	High	Data not collected	Drop the feature	Drop
Other	none	4 duplicate rows	low	recording error	Remove duplicate rows	Remove

Table 1. An overview of data issues

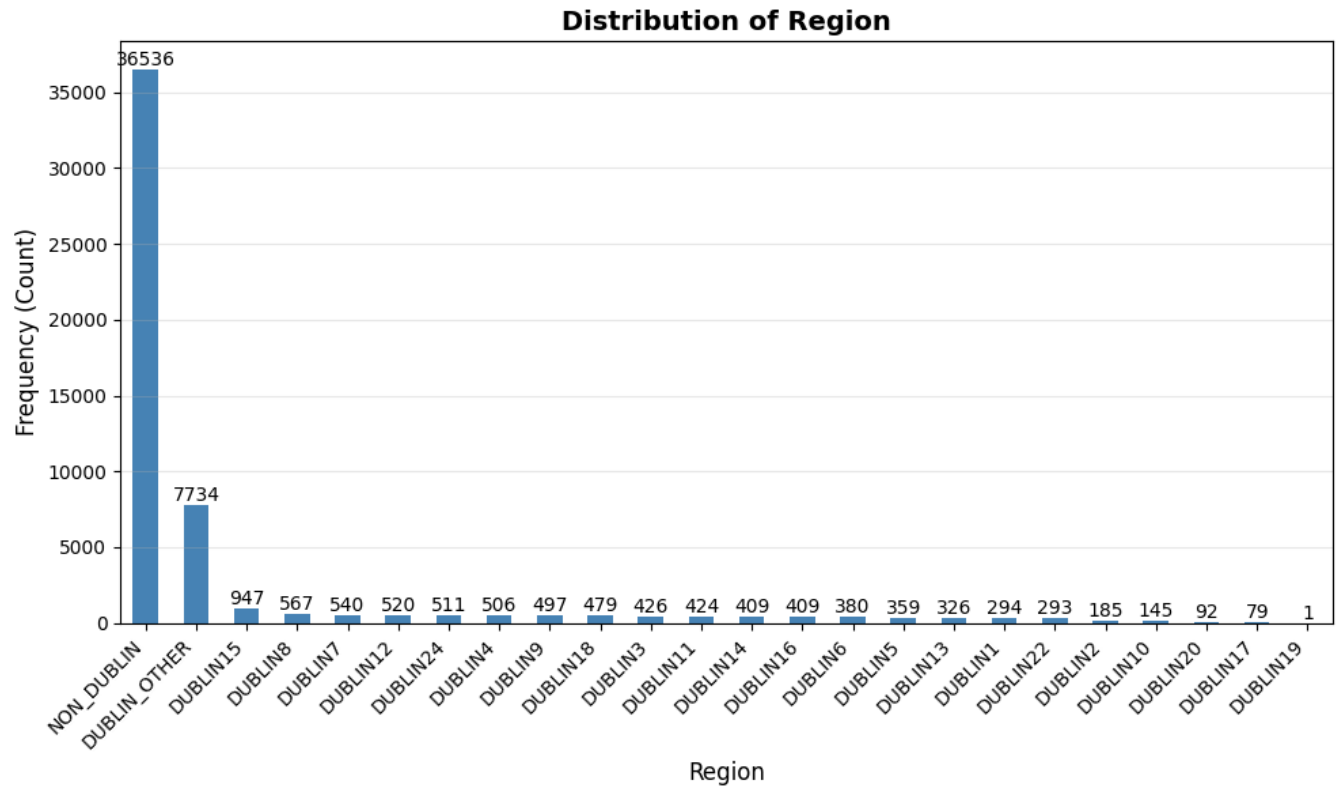


Figure 1. Distribution of *Region*

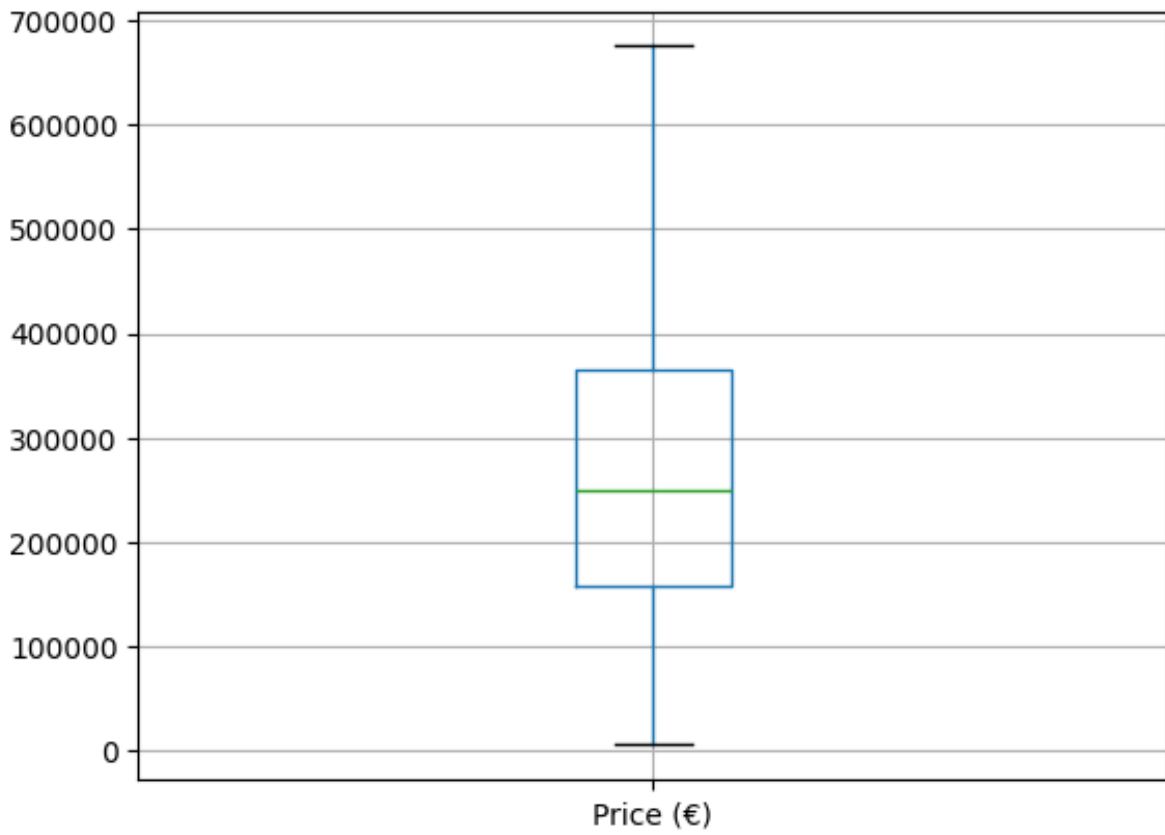


Figure 2. Boxplot after price clamping